

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0946

Ravenna 100 kV
October 15, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0946. The Interconnection Customer proposes a 87.1 MW solar photovoltaic facility interconnecting at the Ravenna 100 kV substation in the Ohio area. The request was received on February 20, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 317 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,391,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ravenna to Cimarron Riv 100.0 kV	Ravenna to El Paso 500.0 kV	206.1	145	8.8
Cimarron River to Cawelo C 100.0 kV	El Paso to Ravenna 500.0 kV	204.7	131	8.8
Cranbrook to Ravenna 100.0 kV	El Paso Substation to Ravenna 500.0 kV	144.7	135	6.5
Manta to Cawel C 100.0 kV line	El Paso Substation to Ravenna 500.0 kV L	120.9	285	8.7
Treat Substation to Mouton Substation 230.0 kV	El Paso to Ravenna 500.0 kV	108.9	575	6.9
Hampton Rd to Medora 345.0 kV	Fishkill Plains to Simpson Stuart Tap 345.0 kV Line	107.1	1,928	6.2
El Paso Substation 500 kV/345 kV Auto	Ganso to El Paso 500.0 kV	106.7	704	5.2
Treat to Chamberlin 230.0 kV	El Paso to Ravenna 500.0 kV Line	103.4	633	6.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cimarron Riv to Ravenna 100.0 kV line	Rebuild line with higher-capacity conductor	\$3,177,000	100% (\$3,177,000)
Caelo C to Cimarron River Substation 100.0 kV L	Rebuild line with higher-capacity conductor	\$5,407,000	100% (\$5,407,000)
Cranbrook to Ravenna 100.0 kV	Rebuild line with higher-capacity conductor	\$9,279,000	100% (\$9,279,000)
Manta to Cawel C 100.0 kV line	Rebuild line with higher-capacity conductor	\$2,285,000	40% (\$908,000)
Treat Substation to Mouton Substation 230.0 kV	Reconductor line	\$12,459,000	5% (\$570,000)
Hampton Rd to Medora 345.0 kV	Reconductor line	\$8,301,000	1% (\$83,000)
El Paso 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$25,370,000	3% (\$748,000)
Treat to Chamberlin 230.0 kV	Reconductor line	\$4,776,000	5% (\$219,000)
	Total allocated to project		\$20,391,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.